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Your first day as a data professional

- The first hire I make at a new company is The DB person. –CTO
- DB people think differently and act differently than regular engineers –CTO
- Something is very wrong with them.--ME

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Who are you?



Operational



Analytical/Reporting



DBE/Development

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The operational person keeps the database running



Operational DB

- Firefighters, Hero’s, and in demand!
- Keeps production databases available, backed up, and secure
- Watches performance and responds when alerts fire
- Applies patches and changes in agreed maintenance windows
- Often on call, so success means nothing breaks
- Fixes Data issues, Always available
- Customer Service Focused

The analytical person turns stored data into numbers people trust



- AI Models, Reporting systems, front end visibility, and works with C level
- Writes the queries behind reports, dashboards, and analysis
- Models data so each business question has one clear answer
- Checks that totals reconcile with the source systems
- Works closely with the people who use the numbers

Analytical/Reporting

The development DB person designs the database and the changes to it



DBE/Development

- Pipelines, Moving data, Migration of data, processes that clean and prepare data, and never interacts with front end
- Designs tables, keys, and indexes for new features and data flows
- Writes stored procedures, functions and migration scripts
- Builds pipelines that load and transform data
- Reviews code and query plans before changes reach production

Operational work is Chaotic and visible, but on-call is the price



Operational

Benefits

- Clear results: systems stay up and recover on time
- Every system that holds data needs someone to run it
- Standard runbooks and checklists to follow
- A direct route into reliability and cloud roles
- **You will always be needed.**

Disadvantages

- On-call shifts and out-of-hours incidents
- Frequent interruptions that break focus
- Little room to build new systems
- Pressure to stay calm while a system is down

Reporting work has direct impact, but depends on data others own



Analytical/Reporting

Benefits

- Direct impact: the numbers you produce shape decisions
- Visible results that business teams use
- Daily SQL practice on real business questions
- A role that mixes technical work and stakeholder contact

Disadvantages

- Output quality depends on source data you do not control
- **You have to CLEAN Everything!**
- A steady queue of requests with competing deadlines
- Metric definitions often disputed between teams
- Repeat reports can feel routine

Development work builds lasting systems, but mistakes reach every user



DBE/Development

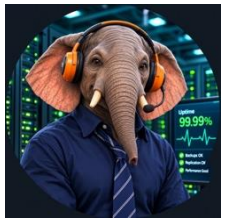
Benefits

- Builds structures that other teams rely on for years
- Deep problem solving on schemas, queries and performance
- A clear path toward senior and architect roles
- Changes are reviewed before they reach production
- **Never chat with customers/end users**

Disadvantages

- A production mistake can affect every user at once
- Schema changes ripple across many applications
- Results can take weeks to show in production
- Needs constant learning across code, tools and platforms
- **Never chat with customers/end users**

Each profile trades a different Pro/Con for a different reward



Operational DB

Pro
Always needed, Critical to operations

Con
ALWAYS NEEDED!!!!



Analytical/Reporting DB

Pro
Highly Visible, well paid, C-Level connected.

Con
Cleaning, cleaning, and more cleaning



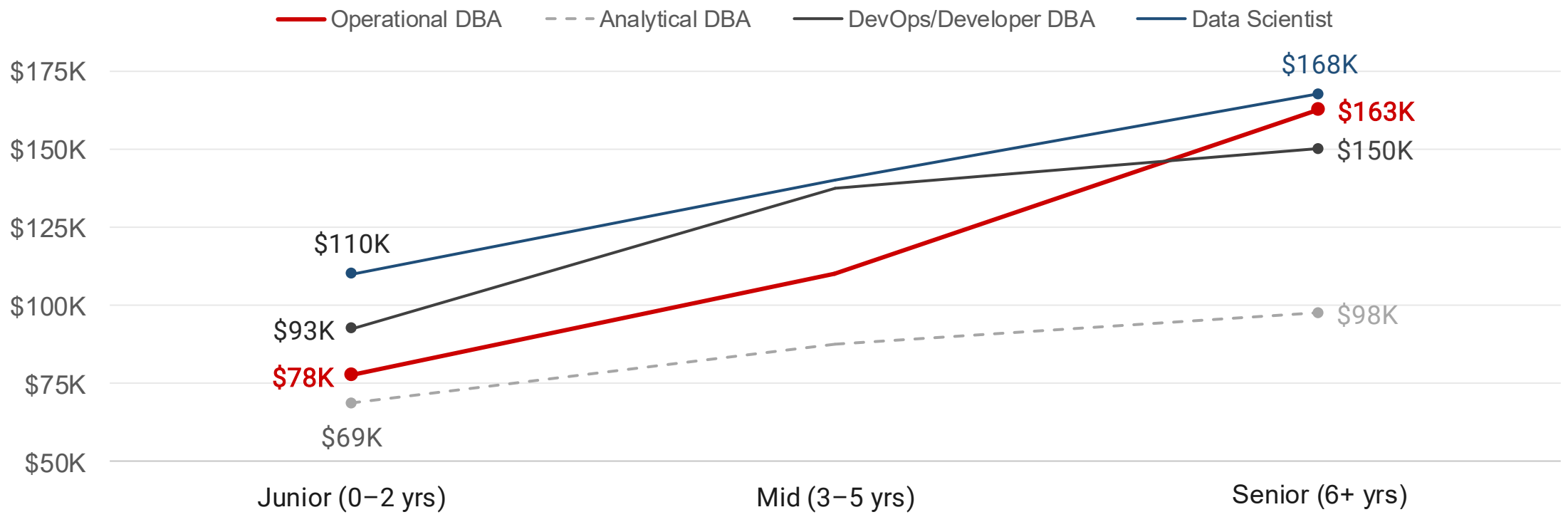
DBE/Development

Pro
Less interaction, more heads-down work.

Con
Never recognized for your work.

Pay by experience level

Operational DBA pay climbs fastest, from about $78K to $163K by senior level



Source: figures supplied by the presenter. Each point is the midpoint of the stated range; Analytical DBA senior is the least certain figure.

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First-day examples

On day one, the operational person learns the system by watching it



Operational

First-day tasks

- Try and find documentation…….you won't find any…..
- Fix a production issue on the first day.
- Start finding the people you need to talk to
- Run a health check: connections, long-running queries, disk space
- Note one manual check you would automate first

On day one, the analytical person checks one report against its source



Analytical/Reporting

First-day tasks

- Find the reporting system and realize that you have to change the whole thing.
- Discuss with stakeholders what they really want.
- Learn that 1 table has all the data you need and it’s in the wrong format.
- Pick one number and trace it back to its source table
- Record every question you could not answer on day one

On day one, Learning path/migration of data and process



DBE/Development

First-day tasks

- Find the current data path/migration on moving data.
- Realize you have to change the whole process.
- Suggest a new tool every week, Databricks, Snowflake, Flink, Spark…..etc…
- Get Dev setup and check something in.

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Focus, study and next steps

What everyone needs to work on

Operational

- Customer Service
- Communication
- Systems Knowledge
- Automation
- Scripting
- SQL

Analytical/Reporting

- Communication
- Data Knowledge/concepts
- Automation
- Python
- Scripting
- SQL

DBE/Development

- DB knowledge
- Transactions
- Automation
- Python
- Pipelines
- Data manipulation
- Scripting
- SQL

What's next?

- Networking
- Networking
- Networking
- Projects
- Projects
- Projects
- SQL

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